Krishi Mitra — Integrated TODO List (Pipeline-Ordered)

Last updated: 2026-08-25 **Sprint goal:** Reach 9/10 prototype score in 2 weeks **Ordering principle:** Pipeline dependency first, then priority. Each task unblocks the next.

Legend

Symbol	Meaning
⭐	Demo-killer —fix first or examiner sees it fail
🔧	Infrastructure —enables other tasks
📊	Evaluation —needs working pipeline + gold labels
📝	Report writing —do last
🆕	NEW —not in original TODO.md
✅	DONE

Sprint Block 1: Foundation (Days 1–2)

These must be done first — everything else depends on them

#	Task	Effort	Status	Original TODO #	Notes
1.1	⭐ Test new Groq model (`openai/gpt-oss-20b` or verified active model)	1 hr	🆕 NEW	#1	Smoke test: `python -c "from llm_client import call_llm; ..."`
1.2	⭐ Copy all 10 corrected files to `backend/`	30 min	🆕 NEW	—	See `CHANGES_INDEX.m` for file list
1.3	🔧 Install deps: `rank_bm25`, `openai`, `indic-nlp-library`	15 min	🆕 NEW	—	One command: `pip install rank_bm25 openai indic-nlp-library`

Table 2 – continued

#	Task	Effort	Status	Original TODO #	Notes
1.4	🔧 Set up `backend/.env` from `.env.example` with real keys	15 min	🆕 NEW	#6	Must have working GROQ_API_K + GENERATION_MODEL
1.5	🔧 **Rebuild Qdrant database with semantic chunking + Indic NLP**	10 min	🆕 NEW	#13	`python vector_db.py` —stop main.py first
1.6	⭐ **Re-label `eval/gold.jsonl`** (chunk IDs changed after semantic chunking)	2–3 hrs	🆕 NEW	–	Use chunk dump command from `CHANGES_SUMMARY` ALL eval numbers are garbage until this is done.

Block 1 exit criteria: LLM responds, database rebuilt, gold labels are fresh.

Sprint Block 2: Demo-Killers (Days 2–5)

These fix the things examiners will notice in the first 2 minutes of your demo

#	Task	Effort	Status	Original TODO #	Notes
2.1	⭐ **Query condensation for multi-turn retrieval**	4–6 hrs	—	#14	**BIGGEST DEMO RISK.** Rewrite follow-up queries using chat history before embedding. Example: “ ?” → “ ?”
2.2	⭐ **Streaming responses (SSE backend + frontend)**	3–4 hrs	—	#15	Groq supports streaming. Frontend needs `EventSource`. Makes 30s feel like 5s.
2.3	⭐ **Expand eval set to n=40–50** from Kisan Call Centre dataset	4–5 hrs	—	#18	n=16 is statistically thin. 40+ is defensible. Mine real Karnataka farmer questions.
2.4	🆕 **Lightweight query expansion** (LLM rewrites vague queries before retrieval)	2 hrs	🆕 NEW	—	Example: “ ?” → “ ”

Table 3 – continued

#	Task	Effort	Status	Original TODO #	Notes
2.5	🆕 **Adversarial test suite** (numeric transposition, injection, empty query)	2 hrs	🆕 NEW	–	Test: 10g→100g hallucination, off-topic injection, PII in query, empty string

Block 2 exit criteria: Follow-up questions work, streaming is live, eval set is 40+, adversarial tests pass.

Sprint Block 3: Evaluation & Metrics (Days 5–7)

These need Block 1 (fresh gold labels) + Block 2 (working pipeline)

#	Task	Effort	Status	Original TODO #	Notes
3.1	📊 **Run threshold sweep** with fresh gold labels	30 min	–	#7	`python eval/threshold_sweep.py` → pick best F1 threshold → update `rag_service.py`
3.2	📊 **Run full retrieval ablation** (dense/sparse/hybrid/BM25/bucketed)	30 min	–	#10	`python eval/run_retrieval_a` → save results for report
3.3	📊 **Run generation eval** with new model + cross-judge	1–2 hrs	–	#2, #4	`python eval/run_eval.py` with `GENERATION_MODEL` set. Use OpenRouter as judge if possible.

Table 4 – continued

#	Task	Effort	Status	Original TODO #	Notes
3.4	📊 **Run refusal test** on unanswerable questions	15 min	—	#11	`python eval/refusal_test.py` → document refusal rate
3.5	📊 **Run numeric faithfulness demo**	5 min	—	#5	`python eval/numeric_faithfuln` → screenshot for report
3.6	📊 **Run metric sanity check** (chrF vs ROUGE-L vs embedding sim)	15 min	—	#3	`python eval/diagnose_metrics.py` → confirm chrF is usable
3.7	📊 **Manual failure-mode review** (read bad answers, classify why)	2 hrs	—	#11	Read low-scoring answers. Categories: retrieval miss, generation hallucination, faithfulness false positive, etc.

Block 3 exit criteria: All eval scripts run, numbers are fresh, threshold is justified, failure modes are classified.

Sprint Block 4: Infrastructure Polish (Days 7–8)

Independent of Blocks 1–3 — can parallel if you have time

#	Task	Effort	Status	Original TODO #	Notes
4.1	🔧 **GitHub Action** running `refusal_test.py` on every push	1 hr	—	#12	One `.github/workflows/ci.yml` file. Engineering maturity signal.
4.2	🔧 **Docker Compose** (backend + frontend + MongoDB + Qdrant)	2–3 hrs	—	#9	One command: `docker-compose up`. Eliminates defense-day "works on my machine" risk.
4.3	🔧 **Pre-flight check script** (verifies .env, DBs, model before startup)	1 hr	🆕 NEW	—	Script checks: MongoDB reachable? Qdrant loaded? LLM responds? Fail fast with clear error.
4.4	🔧 **Query analytics log** (append query + category + score to file)	1 hr	🆕 NEW	—	Simple CSV/ JSONL log. Shows you think about improvement loops.

Block 4 exit criteria: CI passes, Docker works, pre-flight script runs, analytics log exists.

Sprint Block 5: Report Writing (Days 8–10)

Do these last — they need all the numbers and demos from Blocks 1–4

#	Task	Effort	Status	Original TODO #	Notes
5.1	📝 **Document `temperature=0.0` as explicit design choice**	30 min	—	#39	"Deterministic outputs reduce eval variance. Intentional trade-off vs. natural phrasing."

Table 6 – continued

#	Task	Effort	Status	Original TODO #	Notes
5.2	📝 **Document PII/ prompt-injection as out-of-scope**	15 min	—	#41	One sentence: "Input sanitization is deliberately deferred for this prototype."
5.3	📝 **Write "Production Readiness" section**	30 min	—	#42	List auth, rate limiting, logging, CI/ CD, scaling as known deferred gaps. Shows maturity.
5.4	📝 **Write "Future Work" section**	1 hr	🆕 NEW	—	Multi-crop, voice STT/ TTS, domain fine-tuning, real-time prices, farmer feedback loop.
5.5	📝 **Write "Design Choices" section**	1 hr	🆕 NEW	—	Why semantic chunking? Why BGE-M3? Why temperature=0.0? Why hard refusal?
5.6	📝 **Data provenance paragraph**	30 min	—	#8	Source: Karnataka State Agricultural Department + Kisan Call Centre.

Table 6 – continued

#	Task	Effort	Status	Original TODO #	Notes
5.7	📝 **Integrate all eval tables** (retrieval ablation, generation comparison, threshold sweep, refusal accuracy)	2 hrs	🆕 NEW	–	Make tables pretty. Add 95% CI where possible.
5.8	📝 **Final proofread + formatting**	2 hrs	🆕 NEW	–	Consistent terminology, no "morphological chunking" (say "semantic chunking"), check all citations.

Block 5 exit criteria: Report is complete, all numbers are fresh, all design choices are justified.

Sprint Block 6: Defense-Day Prep (Day 10+)

Final polish and dry runs

#	Task	Effort	Status	Original TODO #	Notes
6.1	🆕 **Dry run the live demo** (3–5 preset queries + 1 follow-up)	1 hr	🆕 NEW	–	Practice the flow. Have backup screenshots if live demo fails.

Table 7 – continued

#	Task	Effort	Status	Original TODO #	Notes
6.2	🆕 **Prepare backup slides** (screenshots of streaming, numeric checker, ablation table)	1 hr	🆕 NEW	—	If live demo crashes, you still have evidence.
6.3	🆕 **Prepare 3 "deep dive" talking points**	30 min	🆕 NEW	—	Pick 3 things you' re proud of and can talk for 2 minutes each. Examples: semantic chunking, numeric safety, threshold sweep.
6.4	🆕 **Test on a clean machine** (friend' s laptop, or fresh VM)	2 hrs	🆕 NEW	—	Clone repo, follow your own README, verify it works.

Priority Summary (If You Must Cut)

If you only have 1 week, do this subset:

Priority	Task	Why
P0	Test new Groq model + rebuild DB + re-label gold	Foundation —everything else is garbage without this
P0	Query condensation	**Biggest demo risk** – follow-ups currently fail
P0	Expand eval to n=40	n=16 is not defensible
P1	Streaming	Makes demo feel alive
P1	Run all eval scripts with fresh data	Numbers for report
P1	Adversarial test suite	Shows safety awareness

Table 8 – continued

Priority	Task	Why
P2	GitHub Action	Engineering maturity signal
P2	Docker	Defense-day reliability
P2	Report sections (Design Choices, Future Work, Production Readiness)	Narrative framing
P3	Query expansion	Nice but not critical
P3	Query analytics log	Nice but not critical
P3	Pre-flight script	Nice but not critical

What Got Removed / Deprioritized

These were in your original TODO but are **lower priority** than the new gaps:

Original TODO #	Task	Why Deprioritized
#32	Voice input/ output (STT/ TTS)	Out of prototype scope. Mention in Future Work.
#33	Multi-crop expansion (ragi, tomato)	Requires new data + re-labeling. Future Work.
#40	Structured extraction for dosages	Track 2 / future work. Regex checker (#5) is sufficient for prototype.

How to Use This File

1. **Copy this into your repo** as `TODO_PIPELINE_ORDERED.md`
2. **Check off items as you complete them** — commit the file with checkmarks
3. **If you get stuck on a task**, mark it with ⚠️ and move to the next — don't block the sprint
4. **Update the "Last updated" date** at the top when you make changes

This ordering ensures you never do work that gets invalidated by a later task.